

QSAID WORK INSTRUCTIONS · CODEBASE COHERENTDEMOWI

Record the task once. Ship the work instructions.

A service that turns a shop-floor recording into an ordered, timestamped set of operator work instructions — each step paired with its own video clip — and streams them to the author as they are written, so a procedure that took an afternoon to document takes minutes to review.

VERSION	DATE	AUTHOR	STATUS
1.0 – draft	26 Aug 2026	Faisal Imam	Awaiting sign-off
PRODUCT AREA	COMPANION DOC	BASELINE	REVIEW BY
Industrial AI / QSAID	HLD Rev 1.0	Repo at HEAD	–

Document control and how to read this

This PRD describes one product at two moments: what the current build already does, and what it must do to be usable outside a demo. Every functional and non-functional requirement therefore carries both a priority and an honest implementation status, so the document doubles as a gap list.

LEGEND

MARKER	MEANING
MUST	Required for the first production release. Absence blocks launch.
SHOULD	Expected at launch; a documented workaround is acceptable for one release.
COULD	Desirable, scheduled only if it does not displace a Must.
BUILT	Implemented in the current codebase and verified by reading it.
PARTIAL	Implemented in part, or implemented in a way that does not yet meet the stated criterion.
GAP	Not implemented.

NUMBERS IN THIS DOCUMENT

Targets marked ▲ are proposals that need sign-off, not measurements — the current build has no instrumentation, so no baseline exists for them yet. Everything stated about current behaviour is read from the code. Business figures attributed to manual authoring effort are labelled as assumptions to validate with the pilot site; they are not observed data.

CONTENTS

01	Problem and opportunity	09	Success metrics
02	Product thesis	10	Dependencies and assumptions
03	Goals and non-goals	11	Risks
04	Users and personas	12	Release plan
05	Key journeys	13	Open decisions
06	Scope	A	Interface summary
07	Functional requirements	B	Event contract
08	Non-functional requirements	C	Glossary

Section 01

Problem and opportunity

Manufacturing work instructions are the contract between a process engineer's intent and what an operator actually does at the bench. Producing them is manual, slow, and structurally unrewarding, so they are written once and then left to rot.

What goes wrong today

- **Authoring is expensive.** An engineer watches a recording, scrubs back and forth, transcribes what the expert said, screenshots the important moments, and writes prose around them. The video already contains the procedure; extracting it is the labour.
- **Expertise is undocumented until someone leaves.** The senior technician who knows the sequence is the documentation. Capture happens reactively, under time pressure, usually after a retirement or a resignation.
- **Text-only SOPs are ambiguous.** "Seat the connector fully" means one thing to the person who wrote it and another to a new hire. Without seeing the action, the operator interprets.
- **Revisions do not happen.** When a fixture or a part changes, re-authoring costs as much as the original. So the document diverges from the line, and operators learn to distrust it.
- **Nothing is queryable.** A procedure locked in a PDF cannot be pushed into an MES work order, an LMS training module, or a quality audit trail.

ASSUMPTION TO VALIDATE IN PILOT

We assume a mid-complexity procedure currently costs a process engineer several hours to document from video, and that a usable machine-generated draft removes the majority of that time while leaving the engineer in control of the final wording. The pilot must measure authoring minutes before and after, and the proportion of generated steps accepted without edit. Until then, the business case is a hypothesis.

Why now

Two capabilities became reliable enough to combine: speech transcription with usable word-level timing, and vision models that describe a physical action in a still frame. Neither alone produces a procedure. Together, aligned on a common timeline, they produce a draft that a human can correct faster than they could write it. The remaining work is not model capability — it is timeline alignment, segmentation, and delivery, which is where this product's engineering sits.

Section 02

Product thesis

Record the task once. The system samples the video, transcribes the narration, describes what the worker's hands are doing, folds both timelines into non-overlapping windows, and writes one instruction per window — each delivered with the clip of the video it came from. The engineer reviews and edits a draft rather than facing a blank page.

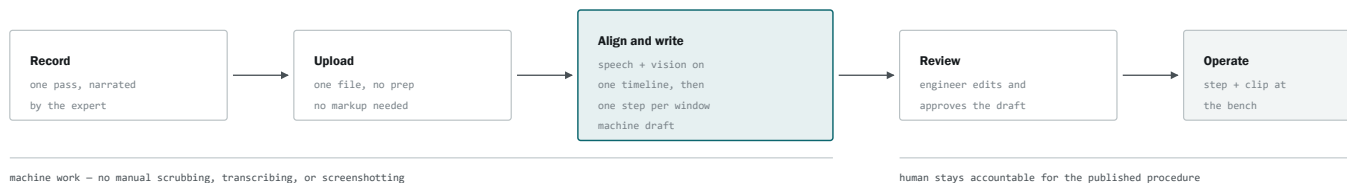


FIG 1 The product removes the extraction labour in the left band. It does not remove the engineer from the right band, and is not intended to — a published work instruction carries safety and quality accountability that must stay with a person.

What makes it defensible

- **Timeline discipline, not prompt cleverness.** Step boundaries are computed deterministically: identical observations collapse, similar neighbours merge, the narrator's own enumeration cues split long windows, gaps are filled, and a final pass guarantees contiguous non-overlapping coverage of the whole video. The same input yields the same boundaries.
- **Evidence per step.** Every instruction ships with the clip it was derived from, so a reviewer verifies in seconds instead of re-scrubbing the source.
- **Streamed, not batched.** Results appear while the video is still processing, which changes the product from a job submission tool into something that feels interactive.
- **API-first.** The UI is one consumer. The same contract lets an MES, an LMS, or a customer's own front end drive the pipeline.

Section 03

Goals and non-goals

PRODUCT GOALS

ID	GOAL	HOW WE WILL KNOW
G-1	Cut authoring effort for a procedure by a large, measured margin	Timed before/after study at the pilot site ▲
G-2	Produce a draft an engineer accepts more often than rewrites	Step acceptance rate without edit $\geq 70\%$ ▲
G-3	Cover the whole recording, with no silently skipped stretches	Sum of step windows equals video duration, no overlaps
G-4	Make progress visible from the first seconds of a job	First step visible well before processing ends
G-5	Be embeddable in existing manufacturing systems	A third party integrates using only the published contract
G-6	Never lose a completed job's output	Every emitted step is durable and replayable after any client loss

EXPLICIT NON-GOALS FOR THIS RELEASE

NOT DOING	WHY
Live camera analysis	The product analyses recordings. Real-time floor monitoring is a different latency and privacy problem.
Editing and approval workflow	v1 hands the draft to the engineer as data and artifacts. Structured review, versioning, and sign-off are the next product, not this one.
Multi-language output	Whisper will transcribe other languages, but instruction quality and terminology have not been validated outside English.
Write-back to MES, PLC, or LMS	Read-only integration first. Writing into systems of record needs a customer-specific validation path.

NOT DOING	WHY
Safety or regulatory certification of output	Generated text is a draft. Accountability stays with the human approver, and the product must say so in the UI.
On-premise or air-gapped deployment	The design depends on hosted model endpoints. An offline variant is a separate architecture.

Section 04

Users and personas

PRIMARY · AUTHOR

Process engineer

Owns the procedure. Documents new lines, changes, and rework loops, and is measured on throughput and defect rate rather than on documentation.

NEEDS a draft that is faithful enough to correct rather than rewrite, and clip evidence to check a step without reopening the source video.

FAILS IF generated steps invent actions, drift from the narration, or arrive as one undifferentiated wall of text.

PRIMARY · CONSUMER

Line operator or trainee

Executes the procedure at a bench, often with gloves on, sometimes on a shared tablet, frequently returning to the same step twice.

NEEDS one step at a time, in order, with the clip for that step and the tools named before the work starts.

FAILS IF steps are out of order, timecodes do not match the clip, or the page assumes a mouse and a quiet room.

SECONDARY · APPROVER

Training or quality lead

Signs off that a published instruction is correct and current, and must answer audit questions about what was published, when, and from what source.

NEEDS the source recording, the transcript, and the generated output preserved together and attributable to one job.

FAILS IF artifacts are transient, unversioned, or unlinked from the video they came from.

SECONDARY · INTEGRATOR

Integration developer

Works for the customer or a systems integrator. Embeds generation into an existing operations application.

NEEDS a stable contract, predictable event ordering, a polling fallback when sockets are blocked by corporate proxies, and honest error messages.

FAILS IF the stream is the only way to get results, or if identifiers and event shapes change between releases.

Section 05

Key journeys

J-1 Author a procedure from a recording

-
- Select.** The engineer picks a recording of the task and chooses sampling density if the default is not right for the video.
 - Start.** The job is created and acknowledged before upload begins, so the progress view opens immediately rather than after the file transfer.
 - Watch.** Upload percentage, then named processing stages, then individual steps appearing one after another in order.
 - Review.** Each step shows its instruction, its start and end timecode, the instruments involved, and a clip cut to exactly that window.
 - Export.** On completion the engineer retrieves the instruction text, the interactive report, and the transcript for downstream use.
-

J-2 Survive a lost connection

-
- Break.** The laptop sleeps or the network drops halfway through a long video.
-

-
- 2 **Continue.** Processing carries on server-side; it is not tied to the viewer.
-
- 3 **Rejoin.** Reconnecting to the same job replays every step already produced, then resumes live delivery with no duplicates and no gaps.
-

J-3 Retry a rejected upload

-
- 1 **Reject.** The file is an unsupported container, is empty, or the transfer is interrupted.
-
- 2 **Explain.** The reason is delivered on the open channel in plain language, and the job returns to an awaiting-upload state rather than failing.
-
- 3 **Retry.** The engineer sends a corrected file against the same job without starting over.
-

J-4 Consume at the bench

-
- 1 **Open.** The operator opens the published procedure on a shared tablet.
-
- 2 **Work.** One step at a time: instruction, tools, clip. Replaying a clip does not lose position in the sequence.
-
- 3 **Verify.** Steps are numbered and timecoded so a supervisor and an operator can refer to the same step unambiguously.
-

Scope

RELEASE BOUNDARIES

BAND	CONTENTS
In — already built	Video ingestion with format validation; parallel frame extraction, transcription, and vision analysis; deterministic timeline segmentation; per-step instruction generation with clip cutting; ordered streaming over WebSocket with durable replay; artifact generation and blob storage; a working reference UI.
In — must be added	Authentication and per-tenant job isolation; job cancellation; durable job execution that survives a restart; job listing and deletion; retention policy; per-stage metrics and cost attribution; scoped storage credentials; artifact links delivered on the live channel.
Out — next release	Structured editing, versioning, and approval; procedure templates and house style; multi-language output; MES/LMS write-back; comparison of two recordings of the same task; operator-facing published view as a first-class product surface.
Out — not planned	Live camera monitoring; worker performance scoring or productivity surveillance; safety certification of generated text; air-gapped deployment.

SCOPE GUARDRAIL

This product describes a physical task from footage of a worker. It must not be positioned or built as a productivity monitoring tool. No requirement in this document identifies, rates, or times an individual, and any future feature that does needs an explicit works-council and privacy review before it is designed, not after.

Section 07

Functional requirements

7.1 Ingestion

ID	REQUIREMENT	PRI	STATUS
FR-101	Accept a single video file per job in common shop-floor containers. <i>AC</i> mp4, mov, avi, mkv, webm, m4v accepted; anything else rejected with a named reason.	MUST	BUILT
FR-102	Return a job identifier before the upload begins, so a client can subscribe first. <i>AC</i> Creation is a metadata-only call and responds without waiting on file bytes.	MUST	BUILT
FR-103	Report upload progress on the same channel the results will arrive on. <i>AC</i> Progress events during transfer, at a frequency that does not flood the channel.	SHOULD	BUILT
FR-104	Accept exactly one upload per job, with a race-safe guard. <i>AC</i> Concurrent uploads to one job: one proceeds, the other is refused with a conflict.	MUST	BUILT
FR-105	Treat a rejected or interrupted upload as retryable, not as a failed job. <i>AC</i> Job returns to awaiting-upload with a message; the same identifier accepts a retry.	MUST	BUILT
FR-106	Enforce and publish a maximum accepted duration and file size. <i>AC</i> Oversize input refused at request time with the limit stated; no partial processing.	MUST	GAP

7.2 Analysis

ID	REQUIREMENT	PRI	STATUS
FR-201	Sample the video at a configurable interval, per job. <i>AC</i> Default one frame per second; interval settable at job creation with a documented floor.	MUST	BUILT
FR-202	Transcribe narration with segment-level timing. <i>AC</i> Transcript returns text plus timed segments usable for alignment.	MUST	BUILT
FR-203	Describe the worker's physical action per sampled frame, ignoring on-screen text. <i>AC</i> Descriptions name hands, tools, and actions; label and document text is not transcribed.	MUST	BUILT
FR-204	Process a silent video without failing. <i>AC</i> No audio track: job completes on visual evidence alone.	MUST	BUILT
FR-205	Avoid paying twice for visually identical frames. <i>AC</i> Near-duplicate consecutive frames reuse the prior description; hits are counted.	SHOULD	PARTIAL
FR-206	Handle recordings longer than a single transcription request can carry. <i>AC</i> Audio is chunked with overlap and segment timings are stitched without drift.	MUST	GAP

7.3 Step generation

ID	REQUIREMENT	PRI	STATUS
FR-301	Produce steps that are contiguous, non-overlapping, and cover the full recording. <i>AC</i> First step starts at zero, last ends at the video duration, no window overlaps its neighbour.	MUST	BUILT
FR-302	Derive step boundaries deterministically from observations, not from a model's discretion. <i>AC</i> Re-running segmentation on identical inputs yields identical boundaries.	MUST	BUILT
FR-303	Split a window where the narrator enumerates sub-actions. <i>AC</i> Enumeration cues in speech produce separate steps instead of one long instruction.	SHOULD	BUILT
FR-304	Keep step length within an operator-readable band. <i>AC</i> Target 8 to 32 seconds; longer windows split, trivial ones merge.	SHOULD	BUILT
FR-305	Ground each instruction only in that window's narration and visible actions. <i>AC</i> Generation receives no other context; output may not reference the video, clip, or timestamps.	MUST	BUILT
FR-306	Preserve the narrator's terminology. <i>AC</i> Domain terms spoken by the expert survive into the instruction rather than being paraphrased away.	MUST	BUILT
FR-307	Never drop a step because generation failed. <i>AC</i> On error, invalid output, or placeholder-grade text, a deterministic writer produces the step from the same evidence.	MUST	BUILT
FR-308	List the instruments and tools relevant to each step. <i>AC</i> Instruments appear as a structured list, sourced from what was seen or heard in that window.	SHOULD	BUILT
FR-309	Suppress duplicate and idle steps without opening coverage holes. <i>AC</i> Near-identical adjacent instructions collapse; removal is refused where it would leave a gap.	SHOULD	BUILT
FR-310	Cut a video clip for every step, bounded exactly by its window. <i>AC</i> One clip per step; a clip that cannot be produced yields a step with no clip rather than a failed job.	MUST	BUILT

7.4 Delivery

ID	REQUIREMENT	PRI	STATUS
FR-401	Stream each step as it is produced, in strict ascending order with no gaps. <i>AC</i> Indices arrive 1, 2, 3 and so on regardless of the order work completed in.	MUST	BUILT
FR-402	Report named processing stages and monotonic progress. <i>AC</i> Progress never moves backwards, even when stages report from parallel workers.	SHOULD	BUILT
FR-403	Replay all produced steps to a reconnecting client, then resume live, with no duplicates. <i>AC</i> A client that misses events receives the full ordered set on reconnect.	MUST	BUILT
FR-404	Provide a request-response fallback for clients that cannot hold a socket. <i>AC</i> A single read returns job state and every persisted step.	MUST	BUILT
FR-405	Keep an idle connection alive through intermediaries. <i>AC</i> A keepalive is emitted after a bounded period of silence.	SHOULD	BUILT
FR-406	Support multiple simultaneous viewers of one job. <i>AC</i> Each viewer receives the full event sequence independently.	COULD	BUILT

ID	REQUIREMENT	PRI	STATUS
FR-407	Deliver artifact links on the live channel when they become available. AC A client that watched a job to completion obtains report and instruction links without a second request.	SHOULD	GAP

7.5 Artifacts and records

ID	REQUIREMENT	PRI	STATUS
FR-501	Persist every emitted step durably as it is emitted. AC A step survives process loss and is retrievable independently of any connection.	MUST	BUILT
FR-502	Produce a consolidated instruction document per job. AC Ordered, timecoded instruction text downloadable as one artifact.	MUST	BUILT
FR-503	Produce a reviewable report pairing instructions with source evidence. AC Self-contained report artifact linking steps to transcript and visual observations.	SHOULD	BUILT
FR-504	Preserve the source recording and transcript alongside the output. AC Source video, plain transcript, timed segments, and subtitle file retained under the job.	MUST	BUILT
FR-505	Serve media to clients without proxying it through the service. AC Clips and artifacts are fetched directly from storage using time-limited read-only links.	SHOULD	BUILT
FR-506	Use one consistent naming convention for stored clips. AC A single documented path pattern; no divergence between code paths.	SHOULD	PARTIAL

7.6 Job management

ID	REQUIREMENT	PRI	STATUS
FR-601	Expose current status, progress, and any error for a job. AC One read returns the full lifecycle state including a human-readable failure reason.	MUST	BUILT
FR-602	Allow a running job to be cancelled. AC Cancellation stops further model spend within seconds and marks the job cancelled, not failed.	MUST	GAP
FR-603	List jobs for the requesting owner, newest first. AC Paged listing scoped to the caller; no cross-tenant visibility.	MUST	GAP
FR-604	Delete a job and every artifact belonging to it. AC One call removes source, clips, steps, artifacts, and metadata; deletion is auditable.	MUST	GAP
FR-605	Resume or fail a job that was interrupted by a service restart. AC No job remains in a running state indefinitely; interrupted work is retried or explicitly failed.	MUST	GAP
FR-606	Expose per-job processing cost. AC Model call counts and estimated cost recorded against each job.	SHOULD	GAP

7.7 Reference user interface

ID	REQUIREMENT	PRI	STATUS
FR-701	Upload, watch progress, and read steps as they arrive. <i>AC</i> Step list grows during processing; each step opens a detail view with its clip.	MUST	BUILT
FR-702	Show service availability and explain an unreachable backend. <i>AC</i> Health is polled; failure states name the likely cause and the fix.	SHOULD	BUILT
FR-703	Expose per-job analysis settings without requiring them. <i>AC</i> Defaults work unattended; sampling and worker settings available to advanced users.	COULD	BUILT
FR-704	State clearly that output is a draft requiring human approval. <i>AC</i> Persistent, non-dismissable notice on any generated procedure view.	MUST	GAP
FR-705	Be operable on a shop-floor tablet. <i>AC</i> Touch targets, legibility at arm's length, and keyboard access meet accessibility guidance.	SHOULD	PARTIAL

Section 08

Non-functional requirements

ID	QUALITY	REQUIREMENT AND TARGET	STATUS
NFR-01	Latency	First step reaches the client within 90 seconds for a five-minute recording at default sampling. ▲	PARTIAL
NFR-02	Throughput	End-to-end processing completes within 1.5× the source duration at default settings. ▲	PARTIAL
NFR-03	Concurrency	At least 5 jobs process simultaneously without stage timeouts or memory exhaustion. ▲	GAP
NFR-04	Scalability	Capacity increases by adding instances; no job state may live in a single process's memory.	GAP
NFR-05	Durability	No emitted step is ever lost, including on abrupt process termination.	BUILT
NFR-06	Recoverability	A job interrupted by a restart is resumed or failed within a bounded window; none stall indefinitely.	GAP
NFR-07	Graceful degradation	A single external failure degrades output quality rather than failing the job, except where no steps can be produced at all.	BUILT
NFR-08	Authentication	Every endpoint requires an authenticated caller; job access is scoped to its owner or tenant.	GAP
NFR-09	Credential handling	The service uses a scoped workload identity; no long-lived storage account key is held by the running process.	GAP
NFR-10	Link exposure	Media links are read-only, short-lived, and revocable without rotating account-wide credentials.	PARTIAL
NFR-11	Origin control	Cross-origin access is restricted to configured origins per environment; wildcard is refused outside local development.	PARTIAL
NFR-12	Retention	Source recordings and derived artifacts are deleted on a configurable schedule, defaulting to a stated maximum.	GAP
NFR-13	Privacy	Footage of identifiable workers is handled under a documented basis; no feature identifies or rates individuals.	GAP
NFR-14	Observability	Per-stage duration, model call counts, cache hit rate, failure rate, and cost per job are emitted as metrics.	GAP
NFR-15	Traceability	Logs are structured and correlated by job identifier across stages and threads.	GAP
NFR-16	Cost control	Cost per minute of processed video is measured, and a per-tenant spend limit can be enforced. ▲	GAP
NFR-17	Determinism	Segmentation is reproducible for identical inputs, so output differences are attributable to generation alone.	BUILT
NFR-18	Portability	Deployable as a single container on any Linux container host; no GPU required.	BUILT
NFR-19	Compatibility	Published contract changes are additive within a major version; event shapes and identifiers stay stable.	PARTIAL
NFR-20	Accessibility	Operator and author surfaces meet WCAG 2.1 AA, including keyboard operation and video captions.	GAP

LAUNCH BLOCKERS

Five gaps stand between the current build and a defensible production release: no authentication (NFR-08), no horizontal scalability (NFR-04), no recovery from restart (NFR-06 , FR-605), no cancellation (FR-602), and no retention or deletion path (NFR-12 , FR-604). None require redesigning the analysis pipeline; all four of the first are infrastructure changes at the edges of the existing system.

Section 09

Success metrics

METRIC	DEFINITION	TARGET ▲	INSTRUMENTED
Authoring time saved	Engineer minutes to a publishable procedure, with the product versus without	≥ 60% reduction	GAP
Step acceptance rate	Share of generated steps published with no wording change	≥ 70%	GAP
Coverage integrity	Jobs whose steps tile the full duration with no overlap or gap	100%	PARTIAL
Time to first step	Upload completion to first step delivered	≤ 90 s at 5 min	GAP
Job success rate	Jobs reaching completion without a failed status	≥ 98%	GAP
Clip attachment rate	Steps delivered with a working clip link	≥ 99%	GAP
Cost per video minute	Total model and storage spend divided by processed minutes	to be set in pilot	GAP
Procedure freshness	Share of published procedures revised within 30 days of a line change	≥ 80%	GAP

The instrumented column is the honest headline of this section: the product currently emits no metrics at all, so none of these targets can be evaluated. NFR-14 is therefore a prerequisite for the pilot, not a follow-up to it.

Section 10

Dependencies and assumptions

EXTERNAL DEPENDENCIES

DEPENDENCY	USED FOR	EXPOSURE IF UNAVAILABLE
Hosted vision and writing model	Per-frame action description; per-step instruction text	Core function stops. Frame analysis aborts after repeated connection failures.
Hosted transcription model	Narration with segment timing	Degrades to visual-only instructions; jobs still complete.
Object storage	System of record for state, steps, clips, artifacts	Total outage. There is no database and no secondary store.
Media tooling in the image	Frame extraction, audio extraction, clip cutting	Steps generate without clips; clips are the product's main evidence, so value drops sharply.
Container host with sockets	Streaming delivery; long-running background work	Falls back to polling; background work suspends if the host idles the process.

ASSUMPTIONS

ID	ASSUMPTION	IF IT PROVES FALSE
A-1	Recordings are narrated by someone who knows the task.	Instruction quality falls to visual description only; value proposition weakens.
A-2	One frame per second is dense enough to catch step transitions.	Sampling must rise, and cost rises proportionally.
A-3	Typical procedures are minutes, not hours, of footage.	Transcription chunking and a durable job queue move from planned to blocking.
A-4	Engineers prefer correcting a draft to writing from scratch.	The whole thesis needs revisiting; test this in pilot before hardening anything.
A-5	Customers accept footage being processed by a hosted model service.	An on-premise or in-tenant architecture becomes a prerequisite, not a later option.
A-6	English-only output is acceptable for the first customers.	Multi-language moves into scope for v1.

Section 11

Risks

RISK	LIKELY	IMPACT	MITIGATION
Generated instructions are plausible but wrong, and someone follows them	Medium	High	Draft framing enforced in product surface (FR-704); human approval required before publication; every step carries its source clip so verification is cheap.
Model spend scales with footage and surprises a customer	High	Medium	Sampling controls already exist; add per-job cost recording (FR-606) and tenant spend limits (NFR-16) before any usage-based commitment.
Single-instance design blocks the first real deployment	High	High	Move fan-out to shared pub/sub and jobs to a durable queue (NFR-04, NFR-06) in the hardening phase; both are contained changes.

RISK	LIKELY	IMPACT	MITIGATION
Long videos fail late, after paying for analysis	Medium	Medium	Enforce documented limits at ingestion (FR-106); chunk transcription (FR-206).
Worker footage triggers privacy or labour objections	Medium	High	Explicit non-goal on monitoring; retention and deletion (NFR-12, FR-604); consent and works-council review as a pilot precondition.
Unauthenticated deployment is reached and abused	Medium	High	Authentication is a launch blocker (NFR-08); until then, restrict network exposure to a known origin and host.
Instruction quality varies by site, camera angle, and lighting	High	Medium	Measure acceptance rate per site; build a recording guidance sheet as part of onboarding rather than tuning prompts per customer.

Section 12

Release plan

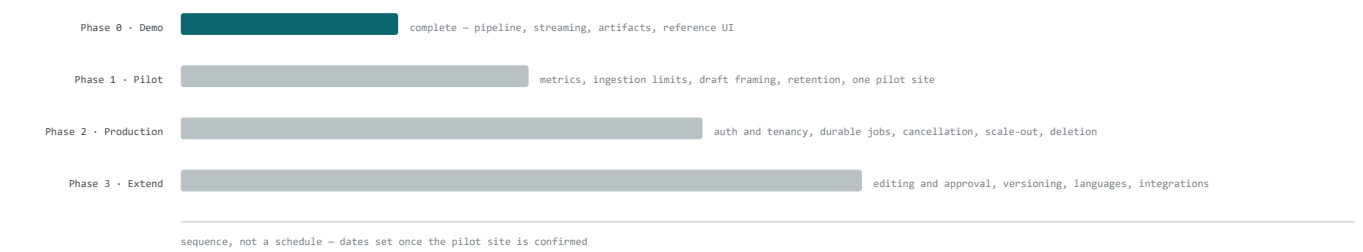


FIG 2 Each phase exits on capability, not on a date. Phase 2 is where the five launch blockers from Section 08 are cleared.

PHASE	CONTENTS	EXIT CRITERIA
0 — Demo DONE	End-to-end generation, ordered streaming, durable steps, artifacts, reference UI	A real factory video produces reviewable, clip-linked steps end to end.
1 — Pilot readiness	NFR-14, NFR-15, FR-106, FR-206, FR-704, NFR-12, FR-604	One customer site runs real procedures; authoring time and acceptance rate are measured, not assumed.
2 — Production	NFR-08, NFR-09, NFR-04, NFR-06, FR-602, FR-603, FR-605, NFR-20	Multi-tenant, authenticated, horizontally scalable, no job stalls, cancellable, accessible.
3 — Extend	Editing and approval workflow, versioning, multi-language, MES and LMS integration	The product owns the whole procedure lifecycle, not just the first draft.

Section 13

Open decisions

ID	DECISION NEEDED	OWNER	NEEDED BY
D-1	Maximum supported video duration and file size, published as a product limit	Product + Engineering	Before Phase 1
D-2	Tenancy model: one deployment per customer, or shared multi-tenant with row-level scoping	Architecture	Before Phase 2
D-3	Identity provider and whether customer SSO is required at launch	Security	Before Phase 2
D-4	Default retention period for source footage, and who may delete	Legal + Product	Before Phase 1

ID	DECISION NEEDED	OWNER	NEEDED BY
D-5	Whether the operator-facing view is our product surface or the customer's	Product	Before Phase 3
D-6	Pricing basis: per video minute, per procedure, or per seat	Commercial	Before Phase 2
D-7	Whether an in-tenant or on-premise deployment must be offered	Product + Architecture	Before Phase 2
D-8	Pilot site, procedures in scope, and who signs off on generated output	Product	Immediately

Appendix A

Interface summary

Behaviour as currently implemented, for reference. The contract is the integration surface referenced by [G-5](#) and [NFR-19](#).

VERB	PATH	PURPOSE
GET	/health	Liveness and service identity
POST	/api/v1/jobs	Create a job from metadata; returns an identifier immediately
POST	/api/v1/jobs/{id}/video	Upload the recording for an existing job; one upload per job
POST	/api/v1/jobs/upload	Legacy combined create and upload; identifier only after full transfer
GET	/api/v1/jobs/{id}	Job state and every persisted step — the polling fallback
WS	/api/v1/jobs/{id}/stream	Event stream for one job

Appendix B

Event contract

EVENT	CARRIES
status	Stage name and monotonic progress; an optional message when an upload is rejected recoverably
step	Step index, instruction text, start and end timecode, clip link, instruments list
complete	Step count; on a reconnect to a finished job, also the artifact links
error	Human-readable failure reason; terminal
ping	Keepalive after a period of silence

KNOWN CONTRACT WRINKLE

Completion is announced as soon as the last step has streamed, so the live `complete` event carries only a step count — the report and instruction links are produced afterwards. A client that watched the job must re-read the job or reconnect to obtain them. [FR-407](#) exists to remove this wrinkle.

Appendix C

Glossary

TERM	MEANING IN THIS DOCUMENT
Work instruction (WI)	One operator-actionable instruction covering one time window of the recording.
Step	A work instruction plus its timecodes, its clip, and its instruments — the unit delivered to clients.
Segment / window	A contiguous span of the recording's timeline that becomes exactly one step.
Coverage	The property that the union of all step windows equals the full recording, with no overlaps.
Visual sequence	Consolidated per-frame action descriptions, grouped into runs of identical observation.

TERM	MEANING IN THIS DOCUMENT
Artifact	A durable file produced per job: instruction text, report, transcript, subtitles, or structured intermediates.
Job	One recording submitted for processing, and everything derived from it, addressed by a single identifier.
Draft	Generated output before a qualified human has reviewed and approved it. All output is a draft.